To **increase TS (Total Solids)** in **FPT (Feed Preparation Tank / Mixing Tank)**, you need to **reduce water content** and/or **increase solid feed concentration**.

Simple rule: More dry material = Higher TS.

Methods to Increase TS in FPT

1. Add Drier (More Dry) Biomass

- Increase **paddy straw**, **press mud**, **cattle dung solids**, **crop residues**, etc.
- Make sure straw is **chopped (2–5 cm)** → gives better mixing and digestion.

Example:

If TS is 6% and target is 10%, **increase dry straw or solids by ~40–60%** gradually.

2. Reduce Water Addition

- Many plants add extra water during feeding → TS automatically reduces.
- **Cut down dilution water** while preparing slurry.

Rule:

Less water = Higher TS.

3. Use Thick Digestate (Recirculate)

- Instead of adding fresh water, add **thick digestate** from the post-digester/secondary digester.
- This gives:
    - Extra **alkalinity**
    - **Microbial seeding**
    - **Higher TS**

This is one of the **best practical methods**.

4. Dewater the Slurry from FPT

- If slurry is too thin → allow **settling** for 30–60 minutes.
- Decant (remove) clear water from the top.
- TS will automatically increase.

Ideal TS Levels (for Biogas Plants)

System Type	Recommended TS in FPT Feed
Wet Digestion	**8–12% TS**
Semi-dry Digestion	**12–18% TS**
Dry Digestion	18–30% TS

Most **CBG / Biogas plants using paddy straw** target:
9–12% TS in feed.

Important Practical Notes

- If TS is **too high (>14%)** → mixing will become **difficult**, viscosity increases.
- If TS is **too low (<6%)** → gas yield decreases per m^3 digester volume (system becomes inefficient).
- Increase TS **gradually** (not more than **1–2% TS per day**) to avoid **process shock**.

One Line for Report

"TS was increased in FPT by reducing dilution water and adding additional chopped dry biomass, ensuring final slurry TS maintained in the range of 9–12%."